

Homework Exercises 2

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After completing all lab exercises below, zip all your solution files into a single archive with your student ID (e.g. u210023.zip) and upload it to eClass before the deadline.

Note, you do not have to submit other lab materials such as lab slides or examples.

1 Experiments with Logisim

Download and install Logisim application from this link to your computer. Once ready, complete following tasks using Logisim. Save your Logisim file (i.e. .circ file) as **ex1.circ**.

1. Construct circuits given in Figure 1 using Logisim. Determine the output of each of the following circuits, assuming that the upper input is 1 and the lower input is 0. What would be the output when the upper input is 0 and the lower input is 1?
2. Construct a circuit given in Figure 2. What Boolean operation does the circuit compute?

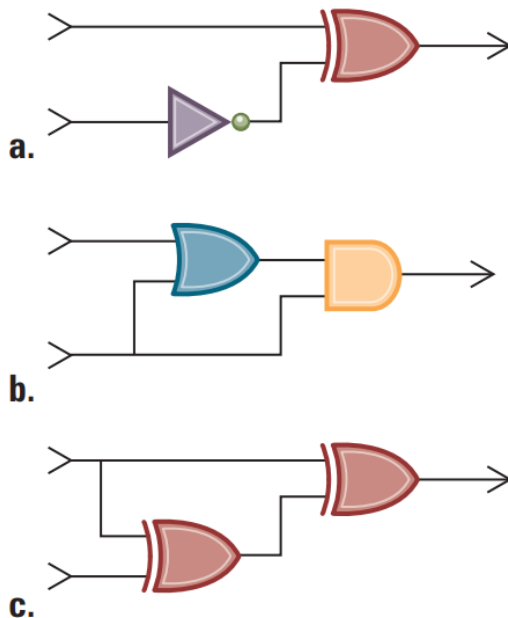


Figure 1: Logical Circuits

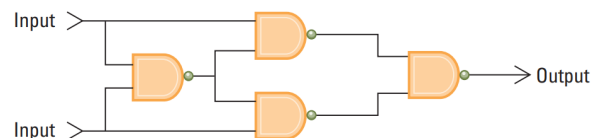


Figure 2: Logical Circuits

2 Python Tasks

Watch **Week 1: Python Basics** lecture videos on this course. Then complete following Python tasks.

1. Write a Python program to get the difference between a given number and 17, if the number is greater than 17 return double the absolute difference. Try code given below.

```
n = 15
if n <= 17:
    print(17 - n)
else:
    print(n - 17) * 2
```

2. Write a Python program to test whether a number is within 100 of 1000 or 2000. Try code given below.

```
n = 1100
isWithin = ((abs(1000 - n) <= 100) or (abs(2000 - n) <= 100))
print(isWithin)
```

3. Write a Python program to print all even numbers from a given numbers list in the same order and stop the printing if any numbers that come after 237 in the sequence.

```
numbers = [
    386, 462, 47, 418, 907, 344, 236, 375, 823, 566, 597, 978, 328, 615, 953, 345,
    399, 162, 758, 219, 918, 237, 412, 566, 826, 248, 866, 950, 626, 949, 687, 217,
    815, 67, 104, 58, 512, 24, 892, 894, 767, 553, 81, 379, 843, 831, 445, 742, 717,
    958, 743, 527
]
```

Write your code by yourself.

4. Write a Python program to compute the *greatest common divisor* (GCD) of two positive integers. Use Euclidean algorithm for that.
5. Write a Python program to get the *least common multiple* (LCM) of two positive integers. Hint, you can use following formula:

$$LCM(a,b) = \frac{|a * b|}{GCD(a,b)}$$